

Multiobjective Optimization

A-posteriori Methods: Concepts and Solution Generation

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A-posteriori Methods - Key Points:

- ① **Weighting Method** - Transform to single objective
- ② **Constraint Method** - Optimize one, constrain others
- ③ Both generate noninferior solutions
- ④ Preferences are articulated **after** generation

Today:

- More advanced generating methods
- Display techniques
- Practical considerations

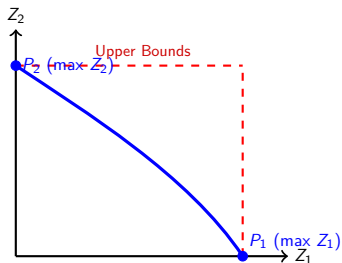
Key Features:

- Iterative method for **approximating** noninferior set
- **Error control** - can specify accuracy
- Efficient for "kinked" or "peaked" noninferior sets

Basic Idea:

- 1 Start with individual optima (endpoints)
- 2 Find point **farthest** from current approximation
- 3 Refine until error criterion met

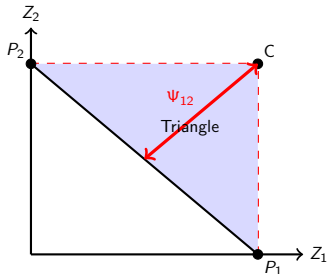
Starting Point



Initial noninferior points: P_1 and P_2 ($Z_1, Z_2 \geq 0$)

Maximum possible error: Triangle area between points

Maximum Possible Error (Ψ_{12})



Triangle: Upper bounds intersect at C

Ψ_{12} = altitude from base P_1P_2

Algorithm Steps

Step 1: Compute weights from slope of $P_i P_{i+1}$

$$w_1 = \frac{Z_2(S_i) - Z_2(S_{i+1})}{Z_1(S_{i+1}) - Z_1(S_i)}$$

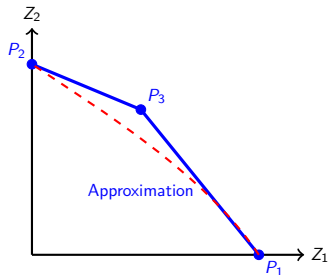
Step 2: Solve weighted problem

Step 3: If new point is above line segment:

- Add to approximation
- Recompute errors

Step 4: Continue until maximum error $\leq$ Tolerance

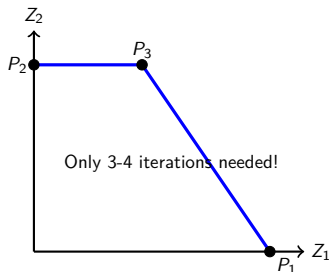
Example Results



Final approximation: $P_1 \rightarrow P_3 \rightarrow P_2$

Accuracy: Controlled by Tolerance T

Efficiency for Elbow Shapes



Advantage: Rapid convergence to "elbow"

Key Features:

- Finds **all** noninferior extreme points
- Based on simplex pivots
- Uses multiobjective simplex tableau

Mathematical Tool:

$$f_j^k = z_j^k - c_j^k$$

Where f_j^k is reduced cost for objective k , variable j

Interpretation: Sign of reduced costs determines noninferiority

Tableau Structure

Basic	a_1	a_2	$\cdots$	a_{n+m}
c^1	c_1^1	c_2^1	$\cdots$	c_{n+m}^1
c^2	c_1^2	c_2^2	$\cdots$	c_{n+m}^2
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
c^p	c_1^p	c_2^p	$\cdots$	c_{n+m}^p
a_{i1}	y_{11}	y_{12}	$\cdots$	$y_{1,n+m}$
a_{i2}	y_{21}	y_{22}	$\cdots$	$y_{2,n+m}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
a_{im}	y_{m1}	y_{m2}	$\cdots$	$y_{m,n+m}$
f^1	f_1^1	f_2^1	$\cdots$	f_{n+m}^1
f^2	f_1^2	f_2^2	$\cdots$	f_{n+m}^2
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
f^p	f_1^p	f_2^p	$\cdots$	f_{n+m}^p

Theorem 1: If $\exists$ nonbasic column a_j with $f_j^k \leq 0$ for all k , then current solution is **inferior**.

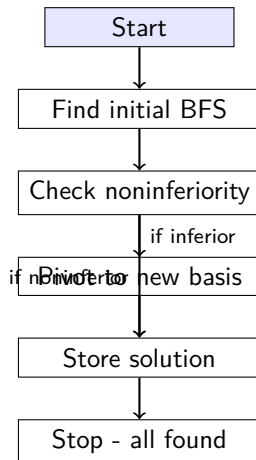
Theorem 2: If $\exists$ nonbasic column a_j with $f_j^k \geq 0$ for all k , then introducing it leads to **inferior** solution.

Theorem 3: If $\exists$ nonbasic columns a_j, a_q with

$$\theta_j f_j^k \leq \theta_q f_q^k \quad \forall k$$

then solution from a_q is dominated by a_j

Flowchart Overview



Application to Sample Problem

Sample problem (from Cohon):

$$Z_1(x_1, x_2) = 5x_1 - 2x_2$$

$$Z_2(x_1, x_2) = -x_1 + 4x_2$$

s.t.

$$-x_1 + x_2 \leq 3, \quad x_1 + x_2 \leq 8$$

$$x_1 \leq 6, \quad x_2 \leq 4, \quad x_1, x_2 \geq 0$$

Noninferior extreme points:

$$B = (6, 0) \rightarrow C = (6, 2) \rightarrow D = (4, 4) \rightarrow E = (1, 4)$$

Sequence of Solutions

Basis	x_1	x_2	Z_1	Z_2
(a_3, a_4, a_5, a_6)	0	0	0	0
(a_2, a_4, a_5, a_6)	0	3	-6	12
(a_2, a_3, a_5, a_1)	1	4	-3	15
(a_2, a_3, a_6, a_1)	4	4	12	12
(a_2, a_4, a_6, a_1)	6	2	26	2
(a_4, a_3, a_6, a_1)	6	0	30	-6

Note: Starts at origin, explores until all noninferior points found

Choosing the Right Tool

Method	Best Use
Weighting	Quick approximation, when weights are meaningful
Constraint	When cuts through set are desired
NISE	Controlled accuracy, "kinked" sets
Multiobjective Simplex	Exact representation, small problems

Consider:

- Computational budget
- Desired accuracy
- Number of objectives

Weighting/Constraint Methods:

r^{p-1} linear programs

NISE Method:

- Fewer solutions than grid
- Controlled by error tolerance

Multiobjective Simplex:

- Number of noninferior extreme points
- Can be 2^n in worst case
- Currently limited to small problems

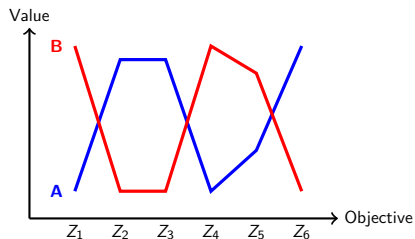
Two-Objective Problems:

- Graphical display in objective space
- Clear and intuitive
- Shows tradeoffs directly

Three or More Objectives:

- Projections onto planes
- Slices (fix other objectives)
- Value paths
- Tabular display

Display for Multiple Objectives



Advantages:

- Concise regardless of objectives
- Shows tradeoffs visually
- Easy to compare alternatives

What We Learned:

- **NISE Method:** Controlled accuracy, efficient for kinked sets
- **Multiobjective Simplex:** Exact, but computationally expensive
- **Choice depends on:** Budget, accuracy, problem size

Key Insight:

"Decision makers like to decide; generating methods allow them to do so." — Cohon (1978)

When Preferences Are Articulated:

- 1 Multiattribute Utility Functions
- 2 Prior Assessment of Weights
- 3 Geometrical Definitions of Best
- 4 Surrogate Worth Tradeoff Method
- 5 Iterative Techniques

Key Question: How much preference information do we need?



COHON, J. L. *Multiobjective Programming and Planning*. Dover Publications, 2013.